Application Narrative:

1. **Funding: if the proposed study has been submitted for funding or is funded, please list funding agency and grant/proposal number, if known. Note that the title of the IRB application should match the title of the grant. If the study is not funded, state N/A**

N/A (faculty startup funds)

2. **Summary:**

Metacognitive Reflection Groups for Improving Student Learning Outcomes in Computing and Programming Courses and Other Courses
This study aims to evaluate the effectiveness of structured metacognitive reflection groups in improving learning outcomes, self-efficacy, professional identity, and professional skills for students in computing, information technology, computer science, and programming courses. Previous research has shown that metacognitive strategies and peer support can improve learning outcomes (Chan et al., 2021; Lim et al., 2022). However, there is a need for scalable, student-centered interventions to support students and increase retention in related programs, which often have high attrition rates.

This study aims to improve learning outcomes for students in computing, information technology, computer science, and programming courses by creating and studying an assignment for structured metacognitive reflection groups. Students will meet regularly in small groups of 3-5 students to discuss their learning experiences, challenges, and strategies across all their university courses, and also fill out weekly reflection questions and set weekly goals, as part of an assignment in the course. We will evaluate how these reflection activities impact students' learning outcomes, self-efficacy, professional identity, and professional skills. The goal is to develop an effective, adoptable intervention to support students and increase retention in related programs, which may also be generalizable to other disciplines.

Our overall research questions are:
1. How does participation in structured metacognitive reflection groups impact students' academic performance in computing-related courses?
2. What changes occur in students' self-efficacy and professional identity as a result of participating in these reflection groups?
3. What effective metacognitive strategies and improvement techniques emerge from the group discussions and student reflections?

Methods:
This research will occur from Fall 2024 to Spring 2029. This study will involve the analysis of data from reflection group assignments for students enrolled in COS100, COS121, COS135, or COS420 at the University of Maine. Starting Spring 2025, students enrolled in COS125, COS140, COS490, and COS498 will also be involved. Starting Fall 2025, students enrolled in NMD100 and HCD101 will also be involved. The reflection groups are a planned part of the course curriculum and will occur regardless of whether <u>the students choose to participate in this research</u>. The research will evaluate the metacognitive reflection group assignments in NMD100 (taught by Joline Blais), HCD101 (taught by Zihan Wu), COS100 (taught by Troy

Schotter and Chris Dufour), COS121, COS125, COS135, COS140 (taught by Troy Schotter) and COS420, COS490, COS498 (taught by: PI Greg Nelson, Laura Gurney, and Troy Schotter). Additionally, starting in Fall 2025, students enrolled in versions of these courses without metacognitive reflection groups, will also be invited to participate in the research, by primarily taking surveys.

For each course with reflection assignments:

1. Students will receive consent forms for participating in the research (subject to / upon approval by the IRB). Students can then opt out of participating in the research, which means their work/assignments from class will not be analyzed/included in the research. The pre- and post-surveys, assignments for the reflection groups and weekly reflection questions will be a part of the course and used for the course and internal evaluation, <u>regardless of whether the students choose to participate in this research.</u>. For example, the pre-survey measures will also be used for an existing common practice in computing courses where students of similar levels are paired to do "pair programming" assignments with each other.
2. In the first week of class (or the first week with IRB approval, whichever is later), administer a pre-survey to assess baseline programming knowledge, other outcomes such as self-efficacy, professional identity, and professional skills, and to gather demographics.
3. In the second week of class (or second week with IRB approval, whichever is later), introduce the concept of reflection groups in a class session.
4. After that, conduct weekly reflection group meetings outside of class time throughout the semester (similar to group project meetings in other classes), where:
    a. Students discuss what's going well in their learning and what is supporting their learning.
    b. Students give updates on their weekly goal from the previous week (if applicable).
    c. Students identify areas for potential improvement in their learning.
    d. Each student chooses one thing to improve for the next week.
5. After each group session, students answer reflection questions as part of the reflection group assignment.
6. Each week, provide a 5-10 minute video on a relevant skill (e.g., help-seeking, problem-solving strategies) as part of readings for the course.
7. About halfway through the semester, hold a larger reflection class activity to present trends, challenges, and successful strategies, and opportunities for students to provide feedback on how to improve the reflection groups.
8. Around the end of the semester, hold a similar larger reflection class activity.
9. Administer a post-survey, including the outcome measures from the pre-survey, and subjective free-response questions to describe the perceived effects of the reflection groups and potential improvements for the groups and the course in general.
10. Longitudinal follow-up: after the completion of the course, students who consented to participate in the research will receive a confidential survey roughly every 6 months on the outcome measures, free-response questions on the impacts of the reflection groups, and what if any practices from the reflection groups they have continued. For example, 8 follow-ups, every six months for 4 years.

For each course without reflection assignments:

1. Students will receive consent forms for participating in the research (subject to / upon approval by the IRB). Students can then opt out of participating in the research, which means their work/assignments from class will not be analyzed/included in the research. The pre- and post-surveys will be a part of the course and used for the course and internal evaluation, regardless of whether the students choose to participate in this research. For example, the pre-survey measures may also be used to match project groups, and help instructors understand students and learning outcomes better.
2. In the first week of class (or the first week with IRB approval, whichever is later), administer the pre-survey, i.e. to assess baseline programming knowledge, other outcomes such as self-efficacy, professional identity, and professional skills, and to gather demographics.
3. Around the end of the semester, administer the post-survey.
4. Longitudinal surveys: after the completion of the course, students who consented to participate in the research will receive a confidential survey roughly every 6 months on the outcome measures, free-response questions on the impacts of the reflection groups, and what if any practices from the reflection groups they have continued. For example, 8 follow-ups, every six months for 4 years.

Data collected:
1.1 Participants' submitted work/assignments: All research participants' work and assignments in the course will be collected for analysis.
1.2 Initial participant pre-survey: Includes demographic data, email addresses, and outcomes data. The questions are in Appendix B; for Fall 25 and on, in Appendix J.
1.3 End of course participant post-survey: Includes email addresses, and outcomes data. The questions are in Appendix C; for Fall 25 and on, in Appendix K.
1.4 Longitudinal follow-up surveys: Includes email addresses, and outcomes data. The questions are in Appendix D; for Fall 25 and on, in Appendix L.
The surveys will be administered through a digital platform Qualtrics. They will each take about 10-15 minutes.

Cecilia.K.Y. Chan and Katherine.K.W. Lee. 2021. Reflection literacy: A multilevel perspective on the challenges of using reflections in higher education through a comprehensive literature review. *Educational Research Review* 32: 100376. https://doi.org/10.1016/j.edurev.2020.100376

Raymond Boon Tar Lim, Kenneth Wee Beng Hoe, and Huili Zheng. 2022. A Systematic Review of the Outcomes, Level, Facilitators, and Barriers to Deep Self-Reflection in Public Health Higher Education: Meta-Analysis and Meta-Synthesis. *Frontiers in Education* 7: 938224. https://doi.org/10.3389/feduc.2022.938224

3. Personnel:

PI: Dr. Gregory Nelson, Assistant Professor, Dept of Computer Science, CLAS

Roles: Instructor, Data Analysis, Course Design and Development, Recruiting
Experience: 5 years of human-computer and education research with human participants

Troy Schotter, Lecturer and PhD Student, Dept of Computer Science, CLAS
Roles: Instructor, Course Design and Development, Data Analysis, Recruiting
Experience: 1 year of education research with human participants

Chris Dufour, Lecturer, Dept of Computer Science, CLAS
Roles: Instructor, Course Design, Data Analysis, Recruiting, Course Design and Development
Experience: First project involving education research with human participants

4. Participant recruitment:

Participant Population:
The participant population for this study will consist of students in COS100. COS121, COS135, and COS420, and starting Spring 2025 in COS125, COS140, COS490, and COS498, and starting Fall 2025 in NMD100 and HCD101. The expected number of participants will be determined by the class size, which is estimated to be approximately 30-50 students for each course. The age range of participants will typically fall within the college student demographic, typically ranging from 18 to 25 years old. Students younger than 18 will be excluded from the study. The target total participant population per course will be approximately 300-500 students over five years.

Participant Identification and Recruitment Procedures:
Participant identification and recruitment will primarily occur within the course. The instructor will be provided with a in-class recruitment script (Appendix E, and starting in Fall25, for courses without reflection groups, Appendix M), which will briefly explain the purpose and goals of the study, voluntary participation, and encourage student participation. Students are also informed that the instructor will not know who was in the study until after grades are submitted.

5. Informed consent:

Electronic consent forms are provided by the instructor emailing them to the students (Appendix A, for Spring 25 see Appendix S25-A, and for Fall 25 and after see Appendix H.); students may opt out by emailing one of the researchers who is not the instructor of their course with the subject line "Opt out of reflection research in <course number e.g. COS100>".
A consent form will be at the beginning of the longitudinal surveys, see Appendix G and for Spring 25 see Appendix S25-G, and for Fall 25 and after see Appendix I. For the longitudinal surveys, submission of the survey will indicate consent.

6. Confidentiality:

Participants in reflection groups will be asked to respect the confidentiality of their peers and not share personal information discussed in the groups. However, we cannot guarantee complete confidentiality in the reflection group meeting setting. Participants will be informed of this limitation in the consent process.

Initially, all the assignment data will be stored on a University of Maine Google Drive titled "Reflection Groups", and in the UMaine Brightspace for the class. Google Drive encrypts all data at rest and in transit with AES-256 bit encryption and is password protected. The pre, post, and follow-up surveys will be stored in Qualtrics.

Within 14 days after the end of a course, or for data collected after the course, within 14 days of collection, all data (including Qualtrics data) will be archived to a separate archival Google Drive folder, with all participant identifiers (such as email addresses) being replaced with a randomly generated encrypted key. During this archiving all data from students not consenting to participate in the research will be filtered out and deleted, by looking up their email addresses and removing the rows of data associated with their email address. The key file that maps to participants' email addresses and participant demographic data will be stored in a separate Google Drive folder. After 5 years from the students' participation in the course, their email address will be deleted from the key file; for example, for students in the Fall 2024 semester, the key will be deleted by end of December 2029. This is to allow students to withdraw their consent to participating in the research later if they wish. We can only delete participant data until the key is deleted.

Work from assignments will be copied onto the archival UMaine Google Drive folder, with any participant identifiers replaced with their encrypted key. The archived data will be retained indefinitely.

Survey data in Qualtrics will be copied into the archival UMaine Google Drive and all participant identifiers replaced with their encrypted key. The archived data will be retained indefinitely.

In summary, the following data from students who participate in the research will be retained indefinitely:
1.1 Participants' submitted work/assignments: All research participants' work and assignments in the course will be collected for analysis.
1.2 Initial participant pre-survey: Includes demographic data, email addresses, and outcomes data. The questions are in Appendix B.
1.3 End of course participant post-survey: Includes email addresses, and outcomes data. The questions are in Appendix C.
1.4 Longitudinal follow-up surveys: Includes email addresses, and outcomes data. The questions are in Appendix D.
Participants' privacy and right to withdraw from the study will be respected, so we will delete any of their data upon request by them as long as it is within 5 years from the students' participation in the course, as otherwise their email address will be deleted from the key file.

7. Risks to participants:

There are risks of time and inconvenience to participants. There is a minimal risk that there will be a data breach. We expect there to be a potential risk of embarrassment in case of a data breach. To minimize the potential for this type of risk causing a breach in confidentiality, the students' email will be changed to a randomized unique identifier that can not be traced back to their email address a period no later than 14 days after course completion. This identifier will be stored separately along with participant's demographic data and any other personally identifiable information.

The surveys will make clear in the introduction that any question can be skipped if the user does not feel comfortable answering.

8. Benefits:

There are no direct benefits to participants, but participants may learn and reflect critically about their learning process leading to improved learning, based on the prior research on the effectiveness of reflection and peer support practices (Chan et al., 2021; Lim et al., 2022).

The overall benefits of the research will include improving our understanding of effective interventions for supporting students in information technology, computing, computer science, and programming courses. It may provide insights into scalable, low-cost methods for improving retention and success. The analysis of the reflections and improvement strategies may provide insight into factors affecting student success and retention in computing fields. The findings could inform pedagogical practices in technical education and potentially be applicable to other STEM fields.
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9. Compensation:

No payments or extra credit for participation will be made to participants. Starting in Fall 25, for the longitudinal surveys, participants will be offered a $10 electronic VISA gift card for each longitudinal survey they submit (see Appendix I). This will be sent to them via email within 48 hours of submitting each survey.

Appendix A: Consent form for participation

UNIVERSITY OF MAINE
CONSENT FORM

Metacognitive Reflection Groups for Improving Student Learning Outcomes in Computing and Programming Courses and Other Courses

Researchers:
- Gregory Nelson, Assistant Professor, Computer Science Department, University of Maine (gregory.nelson@maine.edu) (Principal Investigator, Main Contact)
- Chris Dufour, Lecturer, Computer Science Department, University of Maine,(christopher.dufour@maine.edu)
- Troy Schotter, Lecturer, Computer Science Department, University of Maine (troy.schotter@maine.edu)

You are invited to participate in a research project being led by Dr. Gregory Nelson, a faculty member in the Department of Computer Science at the University of Maine. The purpose of the research is to gauge the effectiveness of student reflection groups and reflection assignments for improving your learning, self-efficacy, professional identity, and professional skills. You must be at least 18 years of age to participate, and participating in COS100, COS121, COS135, or COS420.

What Will You Be Asked to Do?

If you decide to participate, beyond your usual participation in the course, you will be asked to:

- Have your course pre-survey and course post-survey data analyzed by the research team
- Have your assignments and other work for the class analyzed by the research team, including your reflection assignments
- Optionally take confidential longitudinal 10-15 minute follow-up surveys on your learning outcomes and experiences from the reflection groups, for four years (8 total surveys)

The total time commitment is approximately 10-15 minutes for each longitudinal follow-up survey. Beyond that, you will just participate in the class as usual. Participating in the research will not influence your grade. All surveys are confidential.

Risks:

- Time and inconvenience
- Loss of confidentiality of your data, although we will protect your data as listed in the confidentiality section

Benefits:

- There are no direct benefits to you, but you may learn and reflect critically about your learning, self-efficacy, and professional skills more often over time, which may help you better develop skills that can give you a competitive advantage in your education and career

- This research should help us improve our understanding of effective interventions for supporting students, which will benefit future students. The analysis of the reflections and improvement strategies may provide insight into factors affecting student success and retention in computing fields.
- Research such as this may help the academic community adopt better pedagogical practices and potentially be applicable to improving learning in other STEM fields.

Confidentiality

Participants in reflection groups will be asked to respect the confidentiality of their peers and not share personal information discussed in the groups. However, we cannot guarantee complete confidentiality in the reflection group meeting setting.

All data (survey responses, assignments, follow-up surveys) will be stored in password protected computer systems accessible only to the research team using software that provides additional security (encryption). Your student email will be used as an identifier for the duration of the course so that we may relate answers from one method of information gathering to the others as well as give us the ability to remove any data if you decide later that you want any of your data deleted.

Within 14 days after the end of a course, or for data collected after the course, within 14 days of collection, all data will be archived to a separate archival Google Drive folder, with your identifiers (such as email addresses) being replaced with a randomly generated key using software that provides additional security (encryption). The key file that maps your email addresses and your demographic data will be stored in a separate Google Drive folder. After 5 years from your participation in the course, your email address will be deleted from the key file; for example, for students in the Fall 2024 semester, the key will be deleted by end of December 2029. This is to allow you to withdraw your consent to participating in the research later if you wish. We can only delete your data until the key is deleted.

Work from assignments will be copied onto the archival UMaine Google Drive folder, with any of your identifiers replaced with your key. The archived data will be retained indefinitely.

Survey data in Qualtrics will be copied into the archival UMaine Google Drive and all of your identifiers replaced with your key. The archived data will be retained indefinitely.

Voluntary

Participation is voluntary. You may opt out of the research by emailing one of the researchers who is not the instructor of your course with the subject line "Opt out of reflection research in <course number e.g. COS100>". You may skip any survey questions and stop them at any time. You may revoke consent for any portion of this research study at any time until Dec 15, 2029. If you wish to revoke consent, contact one of the researchers as described above, and we will update your status in our records.

Contact Information

If you have any questions about this study, please contact any of the researchers that are a part of the study: gregory.nelson@maine.edu (Principal Investigator, Main Contact), christopher.dufour@maine.edu, troy.schotter@maine.edu. If you have any questions about your rights as a research participant, please contact the Office of Research Compliance, University of Maine, 207-581-2657 (or e-mail umric@maine.edu).

Appendix B: Initial participant pre-survey

The purpose of this survey is to assess the academic background and interests of students. Please help us learn how to improve this class by answering the following questions. If you have any questions about this survey, please reach out to your course instructor. You don't need to have any prior knowledge of the material in this class to succeed and do well in this class.

Your answers are confidential. You will receive credit for completing the survey. Grades are for submitting the survey, not based on your answers. There are no right or wrong answers. You may skip any questions you are uncomfortable answering.

Your Email:

Majors, Career, Prior Learning

What are some future careers you are considering after college? (Open-ended question)

What is your intended major(s) and/or minor(s)? (Open-ended question)

What year of university are you in? (Select one: Freshman (First Year), Sophomore (Second Year), Junior (Third Year), Senior (Fourth Year), Graduate Student, Other - please specify)

Please share a little bit about yourself, such as some hobbies or other parts of your life important to you. (Open-ended question)

How many hours do you work for pay or at a job, each week, on average? (Open-ended question)

Do you have any prior programming experience? (Yes/No)

If applicable, which programming languages have you worked with? (Open-ended question)

How confident do you feel in your programming abilities? (Scale: 1-7, where 1 is "Not confident at all" and 7 is "Very confident")

Demographics

What is your age?

What is your racial and ethnic identity?
Please select any/all that apply.

- [ ] American Indian or Alaska Native
- [ ] Asian
- [ ] Black or African American
- [ ] Hispanic or Latinx
- [ ] Native Hawaiian or Other Pacific Islander
- [ ] White
- [ ] Prefer not to disclose
- [ ] Prefer to self-describe

If you prefer to self-describe, elaborate here

Short answer text

What is your gender?
Please select any/all that apply.

- [ ] Woman
- [ ] Man
- [ ] Non-binary
- [ ] Prefer not to disclose
- [ ] Prefer to self-describe

If you prefer to self-describe, elaborate here

Short answer text

Did you grow up in a household where at least one adult caretaker (mother, father, other caregiver) had a bachelor's degree or higher (or international equivalent)? Yes / No / Not Sure

Do you have any disabilities or accommodations? (Yes/No/Prefer not to answer)

If applicable, do you have any of the following disabilities or chronic conditions? (Please check all that apply.)

Attention deficit

Autism

Blind or visually impaired

Deaf or hard of hearing

Health-related disability

Learning disability

Mental health condition

Mobility-related disability

Speech-related disability
Prefer not to answer

Other (please specify, optional)

Are you an international student? (Yes/No)

What country did you attend high school in?

What was your high school GPA? Please answer as a number out of the maximum, for example 2.2 out of 4.

Are you a native English speaker? (Yes/No)

How comfortable are you reading, writing, speaking, and listening in English? (Scale: 1-7, where 1 is "Not comfortable at all" and 7 is "Extremely comfortable")

Self-efficacy
Self-efficacy is a measurement about your confidence in your ability to perform a task. In the following sections, we ask you to evaluate your self-efficacy with regards to programming.

This survey was designed to be broadly applicable. Nevertheless, it uses a number of technical terms, which may or may not be familiar to you depending on the languages or paradigms you might have experienced. In this sense, interpret the phrase “function or method” as “function” or “method” depending on your background. If you do not recognize a term, please select “no answer” for that item.

		strongly disagree			50/50			strongly agree	no answer
		1	2	3	4	5	6	7	
1.	I can trace the execution of a program.	☐	☐	☐	☐	☐	☐	☐	☐
2.	I can trace the execution of a program with nested conditionals.	☐	☐	☐	☐	☐	☐	☐	☐
3.	I can trace the execution of a program that uses a list or array.	☐	☐	☐	☐	☐	☐	☐	☐
4.	I can trace the execution of a program that iterates over nested structures such as lists of lists or arrays of arrays.	☐	☐	☐	☐	☐	☐	☐	☐
5.	I can trace the execution of a program that contains function or method calls with arguments and return values.	☐	☐	☐	☐	☐	☐	☐	☐
6.	I can trace the execution of a program that uses the same variable name in different locations.	☐	☐	☐	☐	☐	☐	☐	☐
7.	Given a program, I can explain its purpose in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
8.	I can write a program to solve a problem given the solution to a similar problem.	☐	☐	☐	☐	☐	☐	☐	☐
9.	I can write a program that defines and calls functions or methods.	☐	☐	☐	☐	☐	☐	☐	☐
10.	I can explain the difference between built-in and user-defined types.	☐	☐	☐	☐	☐	☐	☐	☐
11.	I can evaluate boolean expressions I encounter in a program.	☐	☐	☐	☐	☐	☐	☐	☐
12.	I can construct a boolean expression to represent a condition described in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
13.	I can explain the difference between a class and an instance of a class.	☐	☐	☐	☐	☐	☐	☐	☐

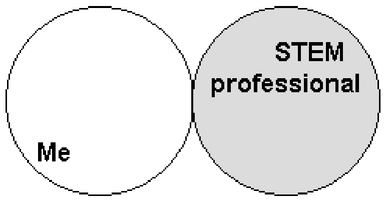
		strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
14.	I can design a set of classes that leverages inheritance where appropriate.	☐	☐	☐	☐	☐	☐	☐	☐
15.	Given a class that has two different functions or methods with the same name, I can identify which one was called.	☐	☐	☐	☐	☐	☐	☐	☐
16.	I can trace the execution of a program that uses recursion.	☐	☐	☐	☐	☐	☐	☐	☐
17.	I can identify recursive patterns in a problem and write a program that exploits this structure.	☐	☐	☐	☐	☐	☐	☐	☐
18.	I can write a function that produces a new list/array by processing data from another list or array.	☐	☐	☐	☐	☐	☐	☐	☐
19.	Given a program that calls functions or methods, I can describe the state of the system at a particular point in execution.	☐	☐	☐	☐	☐	☐	☐	☐
20.	I can decompose a problem into a set of simpler subproblems.	☐	☐	☐	☐	☐	☐	☐	☐
21.	I can use a process to analyze a problem, design a solution, and then implement it in code.	☐	☐	☐	☐	☐	☐	☐	☐
22.	Given the design of a solution and an incorrect program, I can identify the source of the error.	☐	☐	☐	☐	☐	☐	☐	☐
23.	I can design a set of tests that can be used to assess the correctness of a program.	☐	☐	☐	☐	☐	☐	☐	☐
24.	I can correctly trace the execution of a program even if my prior attempts to trace similar programs have not succeeded.	☐	☐	☐	☐	☐	☐	☐	☐
25.	I can write a correctly functioning program even if I have already spent hours fixing errors and have just found a new error.	☐	☐	☐	☐	☐	☐	☐	☐

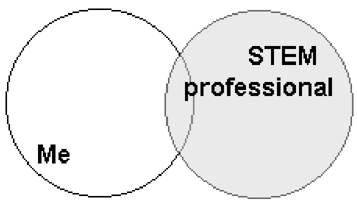
If you have and comments about terms used in this survey, please write them here:

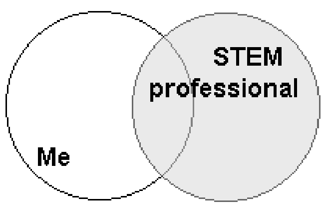
(free response textbox for the question above)

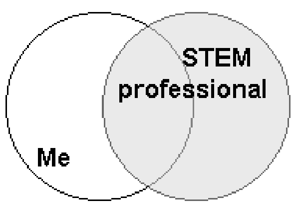
Professional and STEM identity

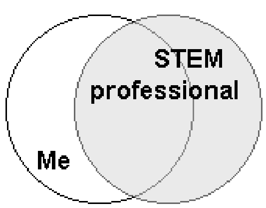
STEM professionals are individuals whose professional activities relate to the STEM fields (Science, Technology, Engineering, or Mathematics). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a STEM professional is.

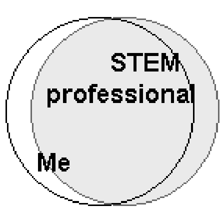












Sense of belonging

1. I feel like I "belong" in computer science.

2. I consider myself as a scientist, technologist, engineer, or mathematician.

3. I take pride in my computing abilities.

on a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree, with a 3 = neutral.

Professional and reflection skills

When I am engaged in the learning process as an INDIVIDUAL:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my effort	○	○	○	○	○	○
I am aware of my thinking	○	○	○	○	○	○
I know my level of motivation	○	○	○	○	○	○
I question my thoughts	○	○	○	○	○	○
I make judgments about the difficulty of a problem	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my existing knowledge	○	○	○	○	○	○
I assess my understanding	○	○	○	○	○	○
I change my strategy when I need to	○	○	○	○	○	○
I am aware of my level of learning	○	○	○	○	○	○
I search for new strategies when needed	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I apply strategies	○	○	○	○	○	○
I assess how I approach the problem	○	○	○	○	○	○
I assess my strategies	○	○	○	○	○	○

When I am engaged in the learning process as a member of a GROUP:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I pay attention to the ideas of others	○	○	○	○	○	○
I listen to the comments of others	○	○	○	○	○	○
I consider the feedback of others	○	○	○	○	○	○
I reflect upon the comments of others	○	○	○	○	○	○
I observe the strategies of others	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I observe how others are doing	○	○	○	○	○	○
I look for confirmation of my understanding from others	○	○	○	○	○	○
I request information from others	○	○	○	○	○	○
I respond to the contributions that others make	○	○	○	○	○	○
I challenge the strategies of others	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I challenge the perspectives of others	○	○	○	○	○	○
I help the learning of others	○	○	○	○	○	○
I monitor the learning of others	○	○	○	○	○	○

I feel a sense of belonging in this class.

Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree

Please describe your answer:

Appendix C: End of course participant post-survey

Please help us learn how to improve this class by answering the following questions. If you have any questions about this survey, please reach out to your course instructor.

Your answers are confidential. You will receive credit for completing the survey. Grades are for submitting the survey, not based on your answers. There are no right or wrong answers. You may skip any questions you are uncomfortable answering.

Your Email:

Majors, Career, Prior Learning
What are some future careers you are considering after college? (Open-ended question)

What is your intended major(s) and/or minor(s)? (Open-ended question)

How many hours do you work for pay or at a job, each week, on average?

What jobs and/or careers do you currently have, if any? (Open-ended question)

How confident do you feel in your programming abilities? (Scale: 1-7, where 1 is "Not confident at all" and 7 is "Very confident")

How comfortable are you reading, writing, speaking, and listening in English? (Scale: 1-7, where 1 is "Not comfortable at all" and 7 is "Extremely comfortable")

Self-efficacy
Self-efficacy is a measurement about your confidence in your ability to perform a task. In the following sections, we ask you to evaluate your self-efficacy with regards to programming.

This survey was designed to be broadly applicable. Nevertheless, it uses a number of technical terms, which may or may not be familiar to you depending on the languages or paradigms you might have experienced. In this sense, interpret the phrase “function or method” as “function” or “method” depending on your background. If you do not recognize a term, please select “no answer” for that item.

		strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
1.	I can trace the execution of a program.	☐	☐	☐	☐	☐	☐	☐	☐
2.	I can trace the execution of a program with nested conditionals.	☐	☐	☐	☐	☐	☐	☐	☐
3.	I can trace the execution of a program that uses a list or array.	☐	☐	☐	☐	☐	☐	☐	☐
4.	I can trace the execution of a program that iterates over nested structures such as lists of lists or arrays of arrays.	☐	☐	☐	☐	☐	☐	☐	☐
5.	I can trace the execution of a program that contains function or method calls with arguments and return values.	☐	☐	☐	☐	☐	☐	☐	☐
6.	I can trace the execution of a program that uses the same variable name in different locations.	☐	☐	☐	☐	☐	☐	☐	☐
7.	Given a program, I can explain its purpose in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
8.	I can write a program to solve a problem given the solution to a similar problem.	☐	☐	☐	☐	☐	☐	☐	☐
9.	I can write a program that defines and calls functions or methods.	☐	☐	☐	☐	☐	☐	☐	☐
10.	I can explain the difference between built-in and user-defined types.	☐	☐	☐	☐	☐	☐	☐	☐
11.	I can evaluate boolean expressions I encounter in a program.	☐	☐	☐	☐	☐	☐	☐	☐
12.	I can construct a boolean expression to represent a condition described in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
13.	I can explain the difference between a class and an instance of a class.	☐	☐	☐	☐	☐	☐	☐	☐

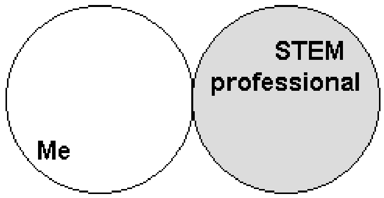
		strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
14.	I can design a set of classes that leverages inheritance where appropriate.	☐	☐	☐	☐	☐	☐	☐	☐
15.	Given a class that has two different functions or methods with the same name, I can identify which one was called.	☐	☐	☐	☐	☐	☐	☐	☐
16.	I can trace the execution of a program that uses recursion.	☐	☐	☐	☐	☐	☐	☐	☐
17.	I can identify recursive patterns in a problem and write a program that exploits this structure.	☐	☐	☐	☐	☐	☐	☐	☐
18.	I can write a function that produces a new list/array by processing data from another list or array.	☐	☐	☐	☐	☐	☐	☐	☐
19.	Given a program that calls functions or methods, I can describe the state of the system at a particular point in execution.	☐	☐	☐	☐	☐	☐	☐	☐
20.	I can decompose a problem into a set of simpler subproblems.	☐	☐	☐	☐	☐	☐	☐	☐
21.	I can use a process to analyze a problem, design a solution, and then implement it in code.	☐	☐	☐	☐	☐	☐	☐	☐
22.	Given the design of a solution and an incorrect program, I can identify the source of the error.	☐	☐	☐	☐	☐	☐	☐	☐
23.	I can design a set of tests that can be used to assess the correctness of a program.	☐	☐	☐	☐	☐	☐	☐	☐
24.	I can correctly trace the execution of a program even if my prior attempts to trace similar programs have not succeeded.	☐	☐	☐	☐	☐	☐	☐	☐
25.	I can write a correctly functioning program even if I have already spent hours fixing errors and have just found a new error.	☐	☐	☐	☐	☐	☐	☐	☐

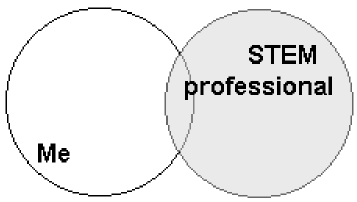
If you have and comments about terms used in this survey, please write them here:

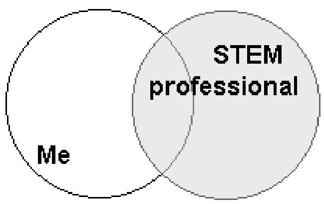
(free response textbox for the question above)

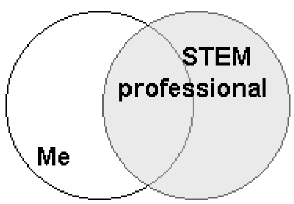
Professional and STEM identity

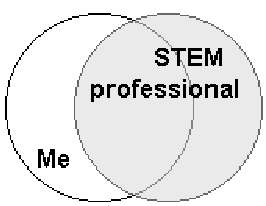
STEM professionals are individuals whose professional activities relate to the STEM fields (Science, Technology, Engineering, or Mathematics). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a STEM professional is.











Sense of belonging

1. I feel like I "belong" in computer science.
2. I consider myself as a scientist, technologist, engineer, or mathematician.
3. I take pride in my computing abilities.

on a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree, with a 3 = neutral.

Professional and reflection skills

When I am engaged in the learning process as an INDIVIDUAL:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my effort						
I am aware of my thinking						
I know my level of motivation						
I question my thoughts						
I make judgments about the difficulty of a problem						
	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my existing knowledge						
I assess my understanding						
I change my strategy when I need to						
I am aware of my level of learning						
I search for new strategies when needed						
	1 - very untrue of me	2	3	4	5	6 - very true of me
I apply strategies						
I assess how I approach the problem						
I assess my strategies						

When I am engaged in the learning process as a member of a GROUP:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I pay attention to the ideas of others	○	○	○	○	○	○
I listen to the comments of others	○	○	○	○	○	○
I consider the feedback of others	○	○	○	○	○	○
I reflect upon the comments of others	○	○	○	○	○	○
I observe the strategies of others	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I observe how others are doing	○	○	○	○	○	○
I look for confirmation of my understanding from others	○	○	○	○	○	○
I request information from others	○	○	○	○	○	○
I respond to the contributions that others make	○	○	○	○	○	○
I challenge the strategies of others	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I challenge the perspectives of others	○	○	○	○	○	○
I help the learning of others	○	○	○	○	○	○
I monitor the learning of others	○	○	○	○	○	○

I feel a sense of belonging in this class.
Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree

Please describe your answer:

Please describe the effects of the weekly reflection groups and reflection questions on yourself:

Please describe the effects of the weekly reflection groups and reflection questions on others in your reflection group:

Please share potential improvements for the weekly reflection groups:

Please share potential improvements for the weekly reflection questions:

Please share potential improvements for the course in general:

Appendix D: Longitudinal follow-up surveys

Please help us learn how to improve the <Course number e.g. COS100> course you took in <Semester Year e.g. Fall 2024>.

Your answers are confidential. There are no right or wrong answers. You may skip any questions you are uncomfortable answering.

Your Email:

Majors, Career

How confident do you feel in your programming abilities? (Scale: 1-7, where 1 is "Not confident at all" and 7 is "Very confident")

How comfortable are you reading, writing, speaking, and listening in English? (Scale: 1-7, where 1 is "Not comfortable at all" and 7 is "Extremely comfortable")

How many hours do you work for pay or at a job, each week, on average?

What jobs and/or careers do you currently have, if any? (Open-ended question)

Part A: If you are still in university or other education/training programs (if not, skip to Part B below):

What are some future careers you are considering after college? (Open-ended question)

What is your intended major(s) and/or minor(s)? (Open-ended question)

What year of university are you in? (Select one: Freshman (First Year), Sophomore (Second Year), Junior (Third Year), Senior (Fourth Year), Graduate Student, Other - please specify)

Part B: If you are no longer taking courses, please answer the following (if not, skip and click the Next button below):

What are some future jobs or careers you are considering after or in addition to your current ones? (Open-ended question)

What major(s) and/or minor(s) did you graduate with and when? (Open-ended question)

Self-efficacy

Self-efficacy is a measurement about your confidence in your ability to perform a task. In the following sections, we ask you to evaluate your self-efficacy with regards to programming.

This survey was designed to be broadly applicable. Nevertheless, it uses a number of technical terms, which may or may not be familiar to you depending on the languages or paradigms you might have experienced. In this sense, interpret the phrase "function or method" as "function" or "method" depending on your background. If you do not recognize a term, please select "no answer" for that item.

		strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
1.	I can trace the execution of a program.	☐	☐	☐	☐	☐	☐	☐	☐
2.	I can trace the execution of a program with nested conditionals.	☐	☐	☐	☐	☐	☐	☐	☐
3.	I can trace the execution of a program that uses a list or array.	☐	☐	☐	☐	☐	☐	☐	☐
4.	I can trace the execution of a program that iterates over nested structures such as lists of lists or arrays of arrays.	☐	☐	☐	☐	☐	☐	☐	☐
5.	I can trace the execution of a program that contains function or method calls with arguments and return values.	☐	☐	☐	☐	☐	☐	☐	☐
6.	I can trace the execution of a program that uses the same variable name in different locations.	☐	☐	☐	☐	☐	☐	☐	☐
7.	Given a program, I can explain its purpose in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
8.	I can write a program to solve a problem given the solution to a similar problem.	☐	☐	☐	☐	☐	☐	☐	☐
9.	I can write a program that defines and calls functions or methods.	☐	☐	☐	☐	☐	☐	☐	☐
10.	I can explain the difference between built-in and user-defined types.	☐	☐	☐	☐	☐	☐	☐	☐
11.	I can evaluate boolean expressions I encounter in a program.	☐	☐	☐	☐	☐	☐	☐	☐
12.	I can construct a boolean expression to represent a condition described in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
13.	I can explain the difference between a class and an instance of a class.	☐	☐	☐	☐	☐	☐	☐	☐

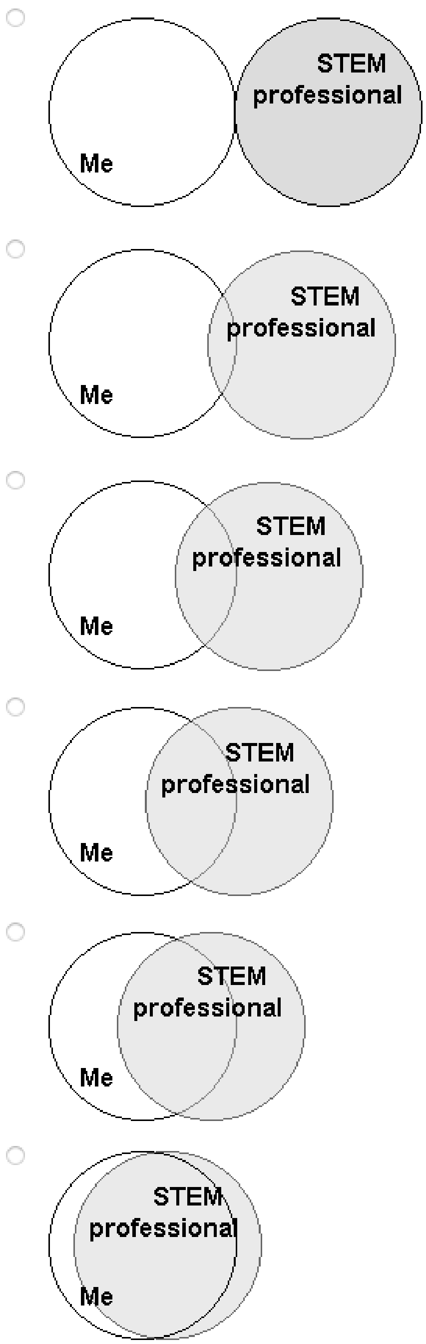
		strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
14.	I can design a set of classes that leverages inheritance where appropriate.	☐	☐	☐	☐	☐	☐	☐	☐
15.	Given a class that has two different functions or methods with the same name, I can identify which one was called.	☐	☐	☐	☐	☐	☐	☐	☐
16.	I can trace the execution of a program that uses recursion.	☐	☐	☐	☐	☐	☐	☐	☐
17.	I can identify recursive patterns in a problem and write a program that exploits this structure.	☐	☐	☐	☐	☐	☐	☐	☐
18.	I can write a function that produces a new list/array by processing data from another list or array.	☐	☐	☐	☐	☐	☐	☐	☐
19.	Given a program that calls functions or methods, I can describe the state of the system at a particular point in execution.	☐	☐	☐	☐	☐	☐	☐	☐
20.	I can decompose a problem into a set of simpler subproblems.	☐	☐	☐	☐	☐	☐	☐	☐
21.	I can use a process to analyze a problem, design a solution, and then implement it in code.	☐	☐	☐	☐	☐	☐	☐	☐
22.	Given the design of a solution and an incorrect program, I can identify the source of the error.	☐	☐	☐	☐	☐	☐	☐	☐
23.	I can design a set of tests that can be used to assess the correctness of a program.	☐	☐	☐	☐	☐	☐	☐	☐
24.	I can correctly trace the execution of a program even if my prior attempts to trace similar programs have not succeeded.	☐	☐	☐	☐	☐	☐	☐	☐
25.	I can write a correctly functioning program even if I have already spent hours fixing errors and have just found a new error.	☐	☐	☐	☐	☐	☐	☐	☐

If you have and comments about terms used in this survey, please write them here:

(free response textbox for the question above)

Professional and STEM identity

STEM professionals are individuals whose professional activities relate to the STEM fields (Science, Technology, Engineering, or Mathematics). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a STEM professional is.



Sense of belonging

1. I feel like I "belong" in computer science.
2. I consider myself as a scientist, technologist, engineer, or mathematician.
3. I take pride in my computing abilities.

on a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree, with a 3 = neutral.

Professional and reflection skills

When I am engaged in the learning process as an INDIVIDUAL:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my effort	○	○	○	○	○	○
I am aware of my thinking	○	○	○	○	○	○
I know my level of motivation	○	○	○	○	○	○
I question my thoughts	○	○	○	○	○	○
I make judgments about the difficulty of a problem	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my existing knowledge	○	○	○	○	○	○
I assess my understanding	○	○	○	○	○	○
I change my strategy when I need to	○	○	○	○	○	○
I am aware of my level of learning	○	○	○	○	○	○
I search for new strategies when needed	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I apply strategies	○	○	○	○	○	○
I assess how I approach the problem	○	○	○	○	○	○
I assess my strategies	○	○	○	○	○	○

When I am engaged in the learning process as a member of a GROUP:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I pay attention to the ideas of others	○	○	○	○	○	○
I listen to the comments of others	○	○	○	○	○	○
I consider the feedback of others	○	○	○	○	○	○
I reflect upon the comments of others	○	○	○	○	○	○
I observe the strategies of others	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I observe how others are doing	○	○	○	○	○	○
I look for confirmation of my understanding from others	○	○	○	○	○	○
I request information from others	○	○	○	○	○	○
I respond to the contributions that others make	○	○	○	○	○	○
I challenge the strategies of others	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I challenge the perspectives of others	○	○	○	○	○	○
I help the learning of others	○	○	○	○	○	○
I monitor the learning of others	○	○	○	○	○	○

I feel a sense of belonging in computer science.
Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree

Please describe your answer:

Please describe the long-term effects on yourself, if any, from the reflection groups and/or weekly reflections from the course?

What practices, if any, have you continued or partly continued from the reflection groups and/or weekly reflections?

If so, around how often do you do each of those practices?

Please share potential improvements for the weekly reflection groups or the course in general:

If there is anything else you would like to add that might be relevant, please do so here.

We value and appreciate your sharing and responses to these questions. Thank you for helping us continue to improve this class, which also helps future students.

Appendix E: Student recruitment script

As a part of this class, you have the opportunity to participate in research on how reflection groups affect your learning. We will be using these reflection groups in the course, and we believe they may help improve your learning outcomes and experiences. Participation in the research is voluntary and will not affect your assignments, class activities, or your grade in any way. I will not know who participated in the study until after grades are submitted.

You can participate by allowing the researchers to analyze your work in the class, including your participation in reflection groups. You may also choose to participate in additional confidential follow-up surveys after the course, which will take about 10-15 minutes each, roughly every six months, for four years (8 surveys total).

More details are in the consent form. We'll take some time now in class to read the consent form and you may follow the directions there to opt-out if you wish, by emailing one of the researchers who is not the instructor of this course, such as <other person on the research team who is not the instructor>. You may decide now or wait until the end of the week. We encourage you to participate as it will help us improve the course, but it is completely optional and will have no effect on your grade. You must be at least 18 years old to participate. If you have any questions, contact <other person on the research team who is not the instructor>, another faculty member involved in the research, whose email is on the consent form and is also here to answer any questions you have.

Appendix F: Other scripts

Sending out surveys:

Greetings,
Here is the link to take the (initial or post or follow-up goes here depending on which survey) survey for the Reflection Groups research project for COS100, COS121, COS135, and COS420: <link to the survey goes here> which will take 10-15 minutes.

Thank you,
<Researcher name>

Sending out longitudinal follow-up surveys:

Greetings,
I hope this message finds you well. I'm reaching out regarding the Reflection Groups research project for <Course name Semester Year e.g. COS100 Fall 2024). The purpose of the research is to gauge the effectiveness of student reflection assignments and groups for improving your learning, self-efficacy, professional identity, and professional skills. We sincerely appreciate your involvement and the valuable insights you've contributed to. <We may add here “For example,” insight or improvement to the course we made from the responses here, about a sentence or two>.

Here is the link to take a confidential longitudinal survey, which will take 10-15 minutes: <link to the survey goes here>

Your ongoing participation helps us gather valuable data and helps us refine and improve how we serve students. If you have any questions or concerns, please don't hesitate to reach out. Overall, you will receive longitudinal surveys roughly every 6 months for 4 years (8 surveys total).

Thank you again for your time and contribution to this research,
<Researcher name>

Appendix G: Longitudinal survey consent form

UNIVERSITY OF MAINE
CONSENT FORM

Metacognitive Reflection Groups for Improving Student Learning Outcomes in Computing and Programming Courses and Other Courses

Researchers:
- Gregory Nelson, Assistant Professor, Computer Science Department, University of Maine (gregory.nelson@maine.edu) (Principal Investigator, Main Contact)
- Chris Dufour, Lecturer, Computer Science Department, University of Maine,(christopher.dufour@maine.edu)
- Troy Schotter, Lecturer, Computer Science Department, University of Maine (troy.schotter@maine.edu)

You are invited to participate in a research project being led by Dr. Gregory Nelson, a faculty member in the Department of Computer Science at the University of Maine. The purpose of the research is to gauge the effectiveness of student reflection groups and reflection assignments for improving your learning, self-efficacy, professional identity, and professional skills. You must be at least 18 years of age to participate, and have participated in COS100, COS121, COS135, or COS420, to participate in this research.

What Will You Be Asked to Do?

If you decide to participate, you will be asked to:

- Take a confidential longitudinal 10-15 minute survey on your learning outcomes and experiences, for four years (8 total surveys)

The total time commitment is approximately 10-15 minutes for each longitudinal survey.

Risks:

- Time and inconvenience
- Loss of confidentiality of your data, although we will protect your data as listed in the confidentiality section

Benefits:

- There are no direct benefits to you, but you may learn and reflect critically about your learning, self-efficacy, and professional skills more often over time, which may help you better develop skills that can give you a competitive advantage in your education and career
- This research should help us improve our understanding of effective interventions for supporting students, which will benefit future students. The analysis of the reflections and improvement strategies may provide insight into factors affecting student success and retention in computing fields.
- Research such as this may help the academic community adopt better pedagogical practices and potentially be applicable to improving learning in other STEM fields.

Confidentiality

Your survey responses will be stored in a password-protected computer using software that provides extra security. Your email address will be used as an identifier to connect your survey response to the data collected while you were enrolled in COS100, COS121, COS135 or COS420.

The key file that maps your email addresses and participant demographic data will be stored in a separate Google Drive folder. After 5 years from your participation in the course, your email address will be deleted from the key file; for example, for students in the Fall 2024 semester, the key will be deleted by end of December 2029. This is to allow you to withdraw your consent to participating in the research later if you wish. We can only delete your data until the key is deleted.

Survey data in Qualtrics will be copied into the archival UMaine Google Drive and all of your identifiers replaced with your key. The archived data will be retained indefinitely.

Voluntary

Participation is voluntary. Submission of this survey indicates consent to participate. You may skip any survey questions and stop them at any time. You may revoke consent for any portion of this research study at any time until Dec 15, 2029. If you wish to revoke consent, you may contact one of the researchers with the subject line "Opt out of reflection research for <course number e.g. COS100", and we will update your status in our records.

Contact Information

If you have any questions about this study, please contact any of the researchers that are a part of the study: gregory.nelson@maine.edu (Principal Investigator, Main Contact), christopher.dufour@maine.edu, troy.schotter@maine.edu. If you have any questions about your rights as a research participant, please contact the Office of Research Compliance, University of Maine, 207-581-2657 (or e-mail umric@maine.edu).

Appendix S25-A: Consent form for participation

UNIVERSITY OF MAINE
CONSENT FORM

Metacognitive Reflection Groups for Improving Student Learning Outcomes in Computing and Programming Courses and Other Courses

Researchers:

- Gregory Nelson, Assistant Professor, Computer Science Department, University of Maine (gregory.nelson@maine.edu) (Principal Investigator, Main Contact)
- Chris Dufour, Lecturer, Computer Science Department, University of Maine,(christopher.dufour@maine.edu)
- Troy Schotter, Lecturer, Computer Science Department, University of Maine (troy.schotter@maine.edu)

You are invited to participate in a research project being led by Dr. Gregory Nelson, a faculty member in the Department of Computer Science at the University of Maine. The purpose of the research is to gauge the effectiveness of student reflection groups and reflection assignments for improving your learning, self-efficacy, professional identity, and professional skills. You must be at least 18 years of age and participating in COS100, COS121, COS125, COS135, COS140, COS420, COS490, or COS498 to participate in this research.

What Will You Be Asked to Do?

If you decide to participate, beyond your usual participation in the course, you will be asked to:

- Have your course pre-survey and course post-survey data analyzed by the research team
- Have your assignments and other work for the class analyzed by the research team, including your reflection assignments
- Optionally take confidential longitudinal 10-15 minute follow-up surveys on your learning outcomes and experiences from the reflection groups, for four years (8 total surveys)

The total time commitment is approximately 10-15 minutes for each longitudinal follow-up survey. Beyond that, you will just participate in the class as usual. Participating in the research will not influence your grade. All surveys are confidential.

Risks:

- Time and inconvenience
- Loss of confidentiality of your data, although we will protect your data as listed in the confidentiality section

Benefits:

- There are no direct benefits to you, but you may learn and reflect critically about your learning, self-efficacy, and professional skills more often over time, which may help you better develop skills that can give you a competitive advantage in your education and career

- This research should help us improve our understanding of effective interventions for supporting students, which will benefit future students. The analysis of the reflections and improvement strategies may provide insight into factors affecting student success and retention in computing fields.
- Research such as this may help the academic community adopt better pedagogical practices and potentially be applicable to improving learning in other STEM fields.

Confidentiality

Participants in reflection groups will be asked to respect the confidentiality of their peers and not share personal information discussed in the groups. However, we cannot guarantee complete confidentiality in the reflection group meeting setting.

All data (survey responses, assignments, follow-up surveys) will be stored in password protected computer systems accessible only to the research team using software that provides additional security (encryption). Your student email will be used as an identifier for the duration of the course so that we may relate answers from one method of information gathering to the others as well as give us the ability to remove any data if you decide later that you want any of your data deleted.

Within 14 days after the end of a course, or for data collected after the course, within 14 days of collection, all data will be archived to a separate archival Google Drive folder, with your identifiers (such as email addresses) being replaced with a randomly generated key using software that provides additional security (encryption). The key file that maps your email addresses and your demographic data will be stored in a separate Google Drive folder. After 5 years from your participation in the course, your email address will be deleted from the key file; for example, for students in the Spring 2025 semester, the key will be deleted by end of May 2030. This is to allow you to withdraw your consent to participating in the research later if you wish. We can only delete your data until the key is deleted.

Work from assignments will be copied onto the archival UMaine Google Drive folder, with any of your identifiers replaced with your key. The archived data will be retained indefinitely.

Survey data in Qualtrics will be copied into the archival UMaine Google Drive and all of your identifiers replaced with your key. The archived data will be retained indefinitely.

Voluntary

Participation is voluntary. You may opt out of the research by emailing one of the researchers who is not the instructor of your course with the subject line "Opt out of reflection research in <course number e.g. COS100>". You may skip any survey questions and stop them at any time. You may revoke consent for any portion of this research study at any time until Dec 15, 2029. If you wish to revoke consent, contact one of the researchers as described above, and we will update your status in our records.

Contact Information

If you have any questions about this study, please contact any of the researchers that are a part of the study: gregory.nelson@maine.edu (Principal Investigator, Main Contact), christopher.dufour@maine.edu, troy.schotter@maine.edu. If you have any questions about your rights as a research participant, please contact the Office of Research Compliance, University of Maine, 207-581-2657 (or e-mail umric@maine.edu).

Appendix S25-G: Longitudinal survey consent form

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Metacognitive Reflection Groups for Improving Student Learning Outcomes in Computing and Programming Courses and Other Courses

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What Will You Be Asked to Do?

If you decide to participate, you will be asked to:
- Take a confidential longitudinal 10-15 minute survey on your learning outcomes and experiences, for four years (8 total surveys)

The total time commitment is approximately 10-15 minutes for each longitudinal survey.

Risks:

- Time and inconvenience
- Loss of confidentiality of your data, although we will protect your data as listed in the confidentiality section

Benefits:
- There are no direct benefits to you, but you may learn and reflect critically about your learning, self-efficacy, and professional skills more often over time, which may help you better develop skills that can give you a competitive advantage in your education and career
- This research should help us improve our understanding of effective interventions for supporting students, which will benefit future students. The analysis of the reflections and improvement strategies may provide insight into factors affecting student success and retention in computing fields.
- Research such as this may help the academic community adopt better pedagogical practices and potentially be applicable to improving learning in other STEM fields.

Confidentiality

Your survey responses will be stored in a password-protected computer using software that provides extra security. Your email address will be used as an identifier to connect your survey response to the data collected while you were enrolled in COS100, COS121, COS125, COS135, COS140, COS420, COS490, or COS498.

The key file that maps your email addresses and participant demographic data will be stored in a separate Google Drive folder. After 5 years from your participation in the course, your email address will be deleted from the key file; for example, for students in the Spring 2025 semester, the key will be deleted by end of May 2030. This is to allow you to withdraw your consent to participating in the research later if you wish. We can only delete your data until the key is deleted.

Survey data in Qualtrics will be copied into the archival UMaine Google Drive and all of your identifiers replaced with your key. The archived data will be retained indefinitely.

Voluntary
Participation is voluntary. Submission of this survey indicates consent to participate. You may skip any survey questions and stop them at any time. You may revoke consent for any portion of this research study at any time until Dec 15, 2029. If you wish to revoke consent, you may contact one of the researchers with the subject line "Opt out of reflection research for <course number e.g. COS100", and we will update your status in our records.

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Appendix H: Consent form for participation

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What Will You Be Asked to Do?

If you decide to participate, beyond your usual participation in the course, you will be asked to:

- Have your course pre-survey and course post-survey data analyzed by the research team
- Have your assignments and other work for the class analyzed by the research team, including any reflection assignment(s)
- Optionally take confidential longitudinal 10-15 minute follow-up surveys on your learning outcomes and experiences, approximately every six months for four years (8 total surveys). For each longitudinal survey you submit, you will receive a $10 VISA gift card.

The total time commitment is approximately 10-15 minutes for each longitudinal follow-up survey. Beyond that, you will just participate in the class as usual. Participating in the research will not influence your grade. All surveys are confidential.

Risks:

- Time and inconvenience
- Loss of confidentiality of your data, although we will protect your data as listed in the confidentiality section

Benefits:

- There are no direct benefits to you, but you may learn and reflect critically about your learning, self-efficacy, and professional skills more often over time, which may help you better develop skills that can give you a competitive advantage in your education and career
- This research should help us improve our understanding of effective teaching and interventions for supporting students, which will benefit future students. The analysis may provide insight into factors affecting student success and retention in computing fields.
- Research such as this may help the academic community adopt better pedagogical practices and potentially be applicable to improving learning in other STEM fields.

Confidentiality

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All data (survey responses, assignments, follow-up surveys) will be stored in password protected computer systems accessible only to the research team using software that provides additional security (encryption). Your student email will be used as an identifier for the duration of the course so that we may relate answers from one method of information gathering to the others as well as give us the ability to remove any data if you decide later that you want any of your data deleted.

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Appendix I: Longitudinal survey consent form

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analysis may provide insight into factors affecting student success and retention in computing fields.

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Survey data in Qualtrics will be copied into the archival UMaine Google Drive and all of your identifiers replaced with your key. The archived data will be retained indefinitely.

Compensation
For each survey, you will be compensated with a $10 electronic VISA gift card. You must reach the end of the survey in order to receive this compensation, though you may skip the occasional question. You will receive your compensation at the conclusion of the survey by email within 48 hours of survey submission.

Voluntary
Participation is voluntary. Submission of this survey indicates consent to participate. You may skip any survey questions and stop them at any time. You may revoke consent for any portion of this research study at any time until Dec 15, 2029. If you wish to revoke consent, you may contact one of the researchers with the subject line "Opt out of reflection research for <course number e.g. COS100", and we will update your status in our records.

Contact Information
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Appendix J: Initial participant pre-survey

The purpose of this survey is to assess the academic background and interests of students. Please help us learn how to improve this class by answering the following questions. If you have any questions about this survey, please reach out to your course instructor. You don't need to have any prior knowledge of the material in this class to succeed and do well in this class.

Your answers are confidential. You will receive credit for completing the survey. Grades are for submitting the survey, not based on your answers. There are no right or wrong answers. You may skip any questions you are uncomfortable answering.

Your Email:

Majors, Career, Prior Learning

What are some future careers you are considering after college? (Open-ended question)

What is your intended major(s) and/or minor(s)? (Open-ended question)

What year of university are you in? (Select one: Freshman (First Year), Sophomore (Second Year), Junior (Third Year), Senior (Fourth Year), Graduate Student, Other - please specify)

Please share a little bit about yourself, such as some hobbies or other parts of your life important to you. (Open-ended question)

How many hours per week do you usually work for pay/job while taking classes? (Open-ended question)

If you are currently employed, what is your profession / job?

Do you have any prior programming experience? (Yes/No)

If applicable, which programming languages have you worked with? (Open-ended question)

How confident do you feel in your programming abilities? (Scale: 1-7, where 1 is "Not confident at all" and 7 is "Very confident")

Demographics

What is your age?

What is your racial and ethnic identity?
Please select any/all that apply.

- ☐ American Indian or Alaska Native
- ☐ Asian
- ☐ Black or African American
- ☐ Hispanic or Latinx
- ☐ Native Hawaiian or Other Pacific Islander
- ☐ White
- ☐ Prefer not to disclose
- ☐ Prefer to self-describe

If you prefer to self-describe, elaborate here

Short answer text

What is your gender?
Please select any/all that apply.

- ☐ Woman
- ☐ Man
- ☐ Non-binary
- ☐ Prefer not to disclose
- ☐ Prefer to self-describe

If you prefer to self-describe, elaborate here

Short answer text

Did you grow up in a household where at least one adult caretaker (mother, father, other caregiver) had a bachelor’s degree or higher (or international equivalent)? Yes / No / Not Sure

Do you have any disabilities or accommodations? (Yes/No/Prefer not to answer)

If applicable, do you have any of the following disabilities or chronic conditions? (Please check all that apply.)

Attention deficit

Autism

Blind or visually impaired

Deaf or hard of hearing

Health-related disability

Learning disability

Mental health condition

Mobility-related disability

Speech-related disability
Prefer not to answer

Other (please specify, optional)

Are you an international student? (Yes/No)

What country did you attend high school in?

What was your high school GPA? Please answer as a number out of the maximum, for example 2.2 out of 4.

Are you a native English speaker? (Yes/No)

How comfortable are you reading, writing, speaking, and listening in English? (Scale: 1-7, where 1 is "Not comfortable at all" and 7 is "Extremely comfortable")

Self-efficacy (Likert Scale)

I will be able to achieve most of the goals that I set for myself.

When facing difficult tasks, I am certain that I will accomplish them.

In general, I think that I can obtain outcomes that are important to me.

I believe I can succeed at most any endeavor to which I set my mind.

I will be able to successfully overcome many challenges.

I am confident that I can perform effectively on many different tasks.

Compared to other people, I can do most tasks very well.

Even when things are tough, I can perform quite well.

I generally manage to solve difficult academic problems if I try hard enough.

I know I can stick to my aims and accomplish my goals in my field of study.

I will remain calm in my exam because I know I will have the knowledge to solve the problems.

I know I can pass the exam if I put in enough work during the semester.

The motto ‘If other people can, I can too’ applies to me when it comes to my field of study.

Self-reflection and Insight Scale (Likert)

I don't often think about my thoughts

I rarely spend time in self-reflection

I frequently examine my feelings

I don't really think about why I behave in the way that I do

I frequently take time to reflect on my thoughts

I often think about the way I feel about things

I am not really interested in analyzing my behavior

It is important for me to evaluate the things that I do

I am very interested in examining what I think about

It is important to me to try to understand what my feelings mean

I have a definite need to understand the way that my mind works

It is important to me to be able to understand how my thoughts arise

I am usually aware of my thoughts

I'm often confused about the way that I really feel about things

I usually have a very clear idea about why I've behaved in a certain way

I'm often aware that I'm having a feeling, but I often don't quite know what it is

My behavior often puzzles me

Thinking about my thoughts makes me more confused

Often I find it difficult to make sense of the way I feel about things

I usually know why I feel the way I do

Lifelong learning skills scale (5-point Likert-type scale (from 1 = never to 5 = always)

I am aware of the areas in which I need to learn.
I am aware of the topics that I am more focused on when learning.
I am aware of my strengths in terms of abilities.
I am aware of what I need to learn.
I am aware of the abilities that I need to improve.

I am able to set my own learning goals.
I am capable of planning my own learning strategies.
I am capable of prioritizing my learning needs.
I am capable of setting up my own learning plan.
I am capable of anticipating my learning outcomes.

I create outlines to assist with my understanding of the material.
I take notes on what I am learning.
During lectures, I ask questions to aid in my learning.
I search for learning resources on my own.
While learning, I record questions that arise.

I monitor my learning progress.
I check if I have achieved my learning goals.
I reflect on my learning process.
I take note of the feedback that my friends give me on my learning.
I make revisions based on the feedback I receive from my friends.

I participate in learning groups that involve interpersonal interactions.
I use communication software to communicate with learning partners.
I express my ideas to learning partners verbally.

I take the initiative to discuss with learning partners.
I maintain a good relationship with learning partners.

Self-efficacy

Self-efficacy is a measurement about your confidence in your ability to perform a task. In the following sections, we ask you to evaluate your self-efficacy with regards to programming.

This survey was designed to be broadly applicable. Nevertheless, it uses a number of technical terms, which may or may not be familiar to you depending on the languages or paradigms you might have experienced. In this sense, interpret the phrase "function or method" as "function" or "method" depending on your background. If you do not recognize a term, please select "no answer" for that item.

		strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
1.	I can trace the execution of a program.	☐	☐	☐	☐	☐	☐	☐	☐
2.	I can trace the execution of a program with nested conditionals.	☐	☐	☐	☐	☐	☐	☐	☐
3.	I can trace the execution of a program that uses a list or array.	☐	☐	☐	☐	☐	☐	☐	☐
4.	I can trace the execution of a program that iterates over nested structures such as lists of lists or arrays of arrays.	☐	☐	☐	☐	☐	☐	☐	☐
5.	I can trace the execution of a program that contains function or method calls with arguments and return values.	☐	☐	☐	☐	☐	☐	☐	☐
6.	I can trace the execution of a program that uses the same variable name in different locations.	☐	☐	☐	☐	☐	☐	☐	☐
7.	Given a program, I can explain its purpose in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
8.	I can write a program to solve a problem given the solution to a similar problem.	☐	☐	☐	☐	☐	☐	☐	☐
9.	I can write a program that defines and calls functions or methods.	☐	☐	☐	☐	☐	☐	☐	☐
10.	I can explain the difference between built-in and user-defined types.	☐	☐	☐	☐	☐	☐	☐	☐
11.	I can evaluate boolean expressions I encounter in a program.	☐	☐	☐	☐	☐	☐	☐	☐
12.	I can construct a boolean expression to represent a condition described in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
13.	I can explain the difference between a class and an instance of a class.	☐	☐	☐	☐	☐	☐	☐	☐

		strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
14.	I can design a set of classes that leverages inheritance where appropriate.	☐	☐	☐	☐	☐	☐	☐	☐
15.	Given a class that has two different functions or methods with the same name, I can identify which one was called.	☐	☐	☐	☐	☐	☐	☐	☐
16.	I can trace the execution of a program that uses recursion.	☐	☐	☐	☐	☐	☐	☐	☐
17.	I can identify recursive patterns in a problem and write a program that exploits this structure.	☐	☐	☐	☐	☐	☐	☐	☐
18.	I can write a function that produces a new list/array by processing data from another list or array.	☐	☐	☐	☐	☐	☐	☐	☐
19.	Given a program that calls functions or methods, I can describe the state of the system at a particular point in execution.	☐	☐	☐	☐	☐	☐	☐	☐
20.	I can decompose a problem into a set of simpler subproblems.	☐	☐	☐	☐	☐	☐	☐	☐
21.	I can use a process to analyze a problem, design a solution, and then implement it in code.	☐	☐	☐	☐	☐	☐	☐	☐
22.	Given the design of a solution and an incorrect program, I can identify the source of the error.	☐	☐	☐	☐	☐	☐	☐	☐
23.	I can design a set of tests that can be used to assess the correctness of a program.	☐	☐	☐	☐	☐	☐	☐	☐
24.	I can correctly trace the execution of a program even if my prior attempts to trace similar programs have not succeeded.	☐	☐	☐	☐	☐	☐	☐	☐
25.	I can write a correctly functioning program even if I have already spent hours fixing errors and have just found a new error.	☐	☐	☐	☐	☐	☐	☐	☐

If you have and comments about terms used in this survey, please write them here:

(free response textbox for the question above)

Professional and STEM identity

(Likert scale)

When I work with technology, I consider whether there are any related ethical issues that I or my team should address.

I feel equipped to reason about ethical issues arising from technology.

It is important for me to be able to effectively communicate about ethical issues arising from technology.

I would like to gain more experience with recognizing and reflecting upon ethical issues arising from technology.

All computer science students should learn about ethical issues arising from technology as part of their computer science program.

When I work with technology in the future, I expect I will more frequently consider whether there are any related ethical issues that I or my team should address.

I routinely think about ethical issues arising from technology.

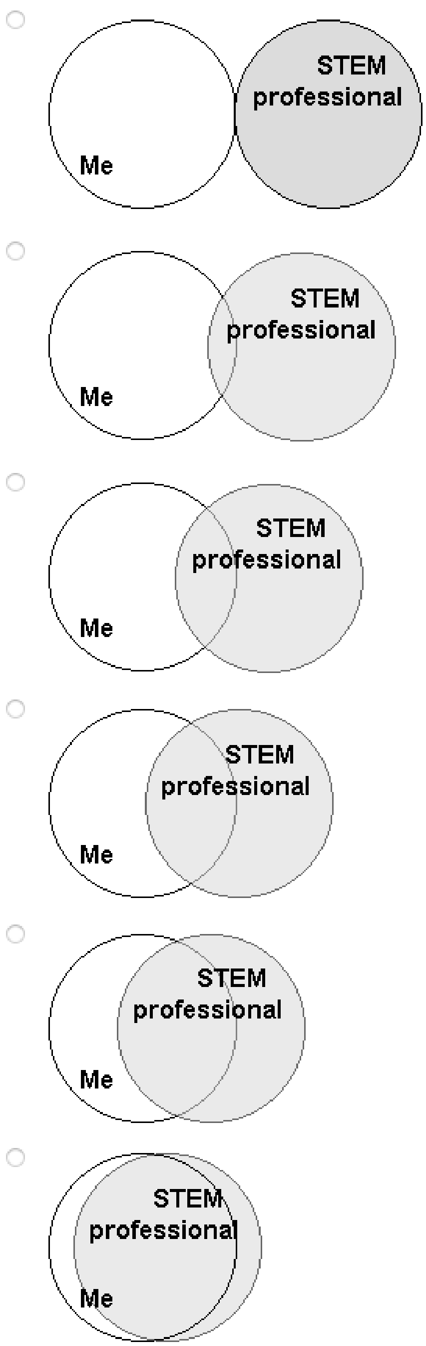
I routinely discuss ethical issues raised by technology with my peers, colleagues, or supervisors.

I am interested in learning more about ethical issues arising from technology.

It is important for computer scientists to consider ethical issues arising from technology in their work.
I feel equipped to identify ethical issues arising from technology.
In the future, I expect I will more frequently reflect upon ethical issues arising from technology.
I have discussed or communicated my position on ethical issues arising technology on many occasions.
I routinely use my ability to recognize, reason about, and discuss ethical issues arising from technology.
I am aware of cases where individuals used their technical expertise to help address the ethical issues arising from technology.
Technology is having an impact on society that raises ethical issues.
I have reflected upon ethical issues arising from technology on many occasions.
I have the skills to carry on a meaningful conversation about ethical issues arising from technology.
I want to improve my ability to recognize, reason about, and discuss the ethical issues arising from technology.
If I am faced with ethical issues arising from technology, I will take steps to address them.
I have noticed ethical issues arising from technology.
When reasoning about ethical issues arising from technology, it is important to consider a wide range of perspectives.
I feel empowered to raise ethical issues arising from technology that may arise in my career.
I expect that, in the future, I will pursue opportunities to improve my ability to recognize, reason about, and discuss ethical issues arising from technology.

I have the capability to take action in response to ethical issues arising from technology, such as suggesting a design modification or requesting that the issue be brought forward for review.

STEM professionals are individuals whose professional activities relate to the STEM fields (Science, Technology, Engineering, or Mathematics). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a STEM professional is.



(another copy of the question above but for "Creative professionals are individuals whose professional activities relate to the creative fields (Art, Design, Creative Writing, Music, Film, Photography, and Games). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a Creative professional is.")

Sense of belonging

1. I feel like I “belong” in computer science.
2. I consider myself as a scientist, technologist, engineer, or mathematician.
3. I take pride in my computing abilities.

on a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree, with a 3 = neutral.

Professional and reflection skills

(Likert scale, 7-point scale from "very inaccurate" to "very accurate.”)
For the teams in this class:

If you make a mistake on this team, it is often held against you.
Members of this team are able to bring up problems and tough issues.
People on this team sometimes reject others for being different.
It is safe to take a risk on this team.
It is difficult to ask other members of this team for help.
No one on this team would deliberately act in a way that undermines my efforts.
Working with members of this team, my unique skills and talents are valued and utilized.

When I am engaged in the learning process as an INDIVIDUAL:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my effort	○	○	○	○	○	○
I am aware of my thinking	○	○	○	○	○	○
I know my level of motivation	○	○	○	○	○	○
I question my thoughts	○	○	○	○	○	○
I make judgments about the difficulty of a problem	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my existing knowledge	○	○	○	○	○	○
I assess my understanding	○	○	○	○	○	○
I change my strategy when I need to	○	○	○	○	○	○
I am aware of my level of learning	○	○	○	○	○	○
I search for new strategies when needed	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I apply strategies	○	○	○	○	○	○
I assess how I approach the problem	○	○	○	○	○	○
I assess my strategies	○	○	○	○	○	○

When I am engaged in the learning process as a member of a GROUP:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I pay attention to the ideas of others						
I listen to the comments of others						
I consider the feedback of others						
I reflect upon the comments of others						
I observe the strategies of others						

	1 - very untrue of me	2	3	4	5	6 - very true of me
I observe how others are doing						
I look for confirmation of my understanding from others						
I request information from others						
I respond to the contributions that others make						
I challenge the strategies of others						

	1 - very untrue of me	2	3	4	5	6 - very true of me
I challenge the perspectives of others						
I help the learning of others						
I monitor the learning of others						

I feel a sense of belonging in this class.

Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree

Please describe your answer:

I feel supported by my (subject i.e. computer science) peers when I face challenges.
Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree

Please describe your answer to "I feel supported by my (subject i.e. computer science) peers when I face challenges."

Appendix K: I End of course participant post-survey

Please help us learn how to improve this class by answering the following questions. If you have any questions about this survey, please reach out to your course instructor.

Your answers are confidential. You will receive credit for completing the survey. Grades are for submitting the survey, not based on your answers. There are no right or wrong answers. You may skip any questions you are uncomfortable answering.

Your Email:

Majors, Career, Prior Learning

What are some future careers you are considering after college? (Open-ended question)

How many hours per week do you usually work for pay/job while taking classes, on average? (Open-ended question)

If you are currently employed, what is your profession / job?

How confident do you feel in your programming abilities? (Scale: 1-7, where 1 is "Not confident at all" and 7 is "Very confident")

How comfortable are you reading, writing, speaking, and listening in English? (Scale: 1-7, where 1 is "Not comfortable at all" and 7 is "Extremely comfortable")

Self-efficacy (Likert Scale)

I will be able to achieve most of the goals that I set for myself.

When facing difficult tasks, I am certain that I will accomplish them.

In general, I think that I can obtain outcomes that are important to me.

I believe I can succeed at most any endeavor to which I set my mind.

I will be able to successfully overcome many challenges.

I am confident that I can perform effectively on many different tasks.

Compared to other people, I can do most tasks very well.

Even when things are tough, I can perform quite well.

I generally manage to solve difficult academic problems if I try hard enough.

I know I can stick to my aims and accomplish my goals in my field of study.

I will remain calm in my exam because I know I will have the knowledge to solve the problems.

I know I can pass the exam if I put in enough work during the semester.

The motto ‘If other people can, I can too’ applies to me when it comes to my field of study.

Self-reflection and Insight Scale (Likert)

I don't often think about my thoughts

I rarely spend time in self-reflection

I frequently examine my feelings

I don't really think about why I behave in the way that I do

I frequently take time to reflect on my thoughts

I often think about the way I feel about things

I am not really interested in analyzing my behavior

It is important for me to evaluate the things that I do

I am very interested in examining what I think about

It is important to me to try to understand what my feelings mean

I have a definite need to understand the way that my mind works

It is important to me to be able to understand how my thoughts arise

I am usually aware of my thoughts

I'm often confused about the way that I really feel about things

I usually have a very clear idea about why I've behaved in a certain way

I'm often aware that I'm having a feeling, but I often don't quite know what it is

My behavior often puzzles me

Thinking about my thoughts makes me more confused

Often I find it difficult to make sense of the way I feel about things

I usually know why I feel the way I do

Lifelong learning skills scale (5-point Likert-type scale (from 1 = never to 5 = always)

I am aware of the areas in which I need to learn.
I am aware of the topics that I am more focused on when learning.
I am aware of my strengths in terms of abilities.
I am aware of what I need to learn.
I am aware of the abilities that I need to improve.

I am able to set my own learning goals.
I am capable of planning my own learning strategies.
I am capable of prioritizing my learning needs.
I am capable of setting up my own learning plan.
I am capable of anticipating my learning outcomes.

I create outlines to assist with my understanding of the material.
I take notes on what I am learning.
During lectures, I ask questions to aid in my learning.
I search for learning resources on my own.
While learning, I record questions that arise.

I monitor my learning progress.
I check if I have achieved my learning goals.
I reflect on my learning process.
I take note of the feedback that my friends give me on my learning.
I make revisions based on the feedback I receive from my friends.

I participate in learning groups that involve interpersonal interactions.
I use communication software to communicate with learning partners.
I express my ideas to learning partners verbally.

I take the initiative to discuss with learning partners.
I maintain a good relationship with learning partners.

Self-efficacy

Self-efficacy is a measurement about your confidence in your ability to perform a task. In the following sections, we ask you to evaluate your self-efficacy with regards to programming.

This survey was designed to be broadly applicable. Nevertheless, it uses a number of technical terms, which may or may not be familiar to you depending on the languages or paradigms you might have experienced. In this sense, interpret the phrase "function or method" as "function" or "method" depending on your background. If you do not recognize a term, please select "no answer" for that item.

		strongly disagree			50/50			strongly agree	no answer
		1	2	3	4	5	6	7	
1.	I can trace the execution of a program.	☐	☐	☐	☐	☐	☐	☐	☐
2.	I can trace the execution of a program with nested conditionals.	☐	☐	☐	☐	☐	☐	☐	☐
3.	I can trace the execution of a program that uses a list or array.	☐	☐	☐	☐	☐	☐	☐	☐
4.	I can trace the execution of a program that iterates over nested structures such as lists of lists or arrays of arrays.	☐	☐	☐	☐	☐	☐	☐	☐
5.	I can trace the execution of a program that contains function or method calls with arguments and return values.	☐	☐	☐	☐	☐	☐	☐	☐
6.	I can trace the execution of a program that uses the same variable name in different locations.	☐	☐	☐	☐	☐	☐	☐	☐
7.	Given a program, I can explain its purpose in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
8.	I can write a program to solve a problem given the solution to a similar problem.	☐	☐	☐	☐	☐	☐	☐	☐
9.	I can write a program that defines and calls functions or methods.	☐	☐	☐	☐	☐	☐	☐	☐
10.	I can explain the difference between built-in and user-defined types.	☐	☐	☐	☐	☐	☐	☐	☐
11.	I can evaluate boolean expressions I encounter in a program.	☐	☐	☐	☐	☐	☐	☐	☐
12.	I can construct a boolean expression to represent a condition described in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
13.	I can explain the difference between a class and an instance of a class.	☐	☐	☐	☐	☐	☐	☐	☐

	strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
14. I can design a set of classes that leverages inheritance where appropriate.	☐	☐	☐	☐	☐	☐	☐	☐
15. Given a class that has two different functions or methods with the same name, I can identify which one was called.	☐	☐	☐	☐	☐	☐	☐	☐
16. I can trace the execution of a program that uses recursion.	☐	☐	☐	☐	☐	☐	☐	☐
17. I can identify recursive patterns in a problem and write a program that exploits this structure.	☐	☐	☐	☐	☐	☐	☐	☐
18. I can write a function that produces a new list/array by processing data from another list or array.	☐	☐	☐	☐	☐	☐	☐	☐
19. Given a program that calls functions or methods, I can describe the state of the system at a particular point in execution.	☐	☐	☐	☐	☐	☐	☐	☐
20. I can decompose a problem into a set of simpler subproblems.	☐	☐	☐	☐	☐	☐	☐	☐
21. I can use a process to analyze a problem, design a solution, and then implement it in code.	☐	☐	☐	☐	☐	☐	☐	☐
22. Given the design of a solution and an incorrect program, I can identify the source of the error.	☐	☐	☐	☐	☐	☐	☐	☐
23. I can design a set of tests that can be used to assess the correctness of a program.	☐	☐	☐	☐	☐	☐	☐	☐
24. I can correctly trace the execution of a program even if my prior attempts to trace similar programs have not succeeded.	☐	☐	☐	☐	☐	☐	☐	☐
25. I can write a correctly functioning program even if I have already spent hours fixing errors and have just found a new error.	☐	☐	☐	☐	☐	☐	☐	☐

If you have and comments about terms used in this survey, please write them here:

(free response textbox for the question above)

Professional and STEM identity

(Likert scale)

When I work with technology, I consider whether there are any related ethical issues that I or my team should address.

I feel equipped to reason about ethical issues arising from technology.

It is important for me to be able to effectively communicate about ethical issues arising from technology.

I would like to gain more experience with recognizing and reflecting upon ethical issues arising from technology.

All computer science students should learn about ethical issues arising from technology as part of their computer science program.

When I work with technology in the future, I expect I will more frequently consider whether there are any related ethical issues that I or my team should address.

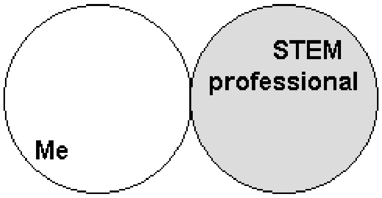
I routinely think about ethical issues arising from technology.

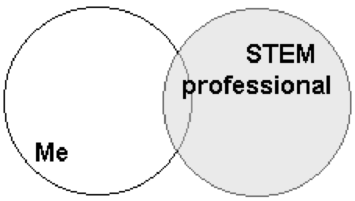
I routinely discuss ethical issues raised by technology with my peers, colleagues, or supervisors.

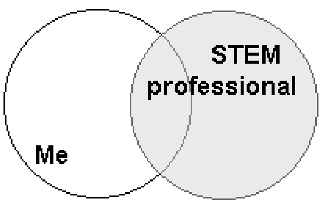
I am interested in learning more about ethical issues arising from technology.
It is important for computer scientists to consider ethical issues arising from technology in their work.
I feel equipped to identify ethical issues arising from technology.
In the future, I expect I will more frequently reflect upon ethical issues arising from technology.
I have discussed or communicated my position on ethical issues arising technology on many occasions.
I routinely use my ability to recognize, reason about, and discuss ethical issues arising from technology.
I am aware of cases where individuals used their technical expertise to help address the ethical issues arising from technology.
Technology is having an impact on society that raises ethical issues.
I have reflected upon ethical issues arising from technology on many occasions.
I have the skills to carry on a meaningful conversation about ethical issues arising from technology.
I want to improve my ability to recognize, reason about, and discuss the ethical issues arising from technology.
If I am faced with ethical issues arising from technology, I will take steps to address them.
I have noticed ethical issues arising from technology.
When reasoning about ethical issues arising from technology, it is important to consider a wide range of perspectives.
I feel empowered to raise ethical issues arising from technology that may arise in my career.
I expect that, in the future, I will pursue opportunities to improve my ability to recognize, reason about, and discuss ethical issues arising from technology.

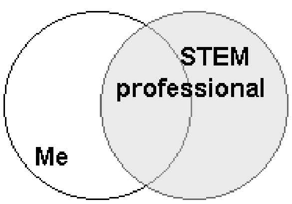
I have the capability to take action in response to ethical issues arising from technology, such as suggesting a design modification or requesting that the issue be brought forward for review.

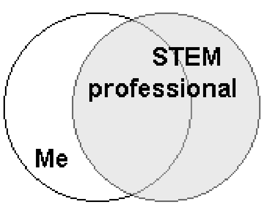
STEM professionals are individuals whose professional activities relate to the STEM fields (Science, Technology, Engineering, or Mathematics). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a STEM professional is.

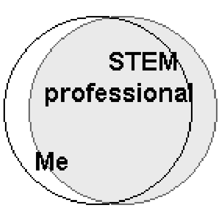












(another copy of the question above but for “Creative professionals are individuals whose professional activities relate to the creative fields (Art, Design, Creative Writing, Music, Film, Photography, and Games). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a Creative professional is.”)

Sense of belonging

1. I feel like I "belong" in computer science.

2. I consider myself as a scientist, technologist, engineer, or mathematician.

3. I take pride in my computing abilities.

on a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree, with a 3 = neutral.

Professional and reflection skills

(Likert scale, 7-point scale from "very inaccurate" to "very accurate.")
For the teams in this class:

If you make a mistake on this team, it is often held against you.
Members of this team are able to bring up problems and tough issues.
People on this team sometimes reject others for being different.
It is safe to take a risk on this team.
It is difficult to ask other members of this team for help.
No one on this team would deliberately act in a way that undermines my efforts.
Working with members of this team, my unique skills and talents are valued and utilized.

When I am engaged in the learning process as an INDIVIDUAL:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my effort	○	○	○	○	○	○
I am aware of my thinking	○	○	○	○	○	○
I know my level of motivation	○	○	○	○	○	○
I question my thoughts	○	○	○	○	○	○
I make judgments about the difficulty of a problem	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my existing knowledge	○	○	○	○	○	○
I assess my understanding	○	○	○	○	○	○
I change my strategy when I need to	○	○	○	○	○	○
I am aware of my level of learning	○	○	○	○	○	○
I search for new strategies when needed	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I apply strategies	○	○	○	○	○	○
I assess how I approach the problem	○	○	○	○	○	○
I assess my strategies	○	○	○	○	○	○

When I am engaged in the learning process as a member of a GROUP:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I pay attention to the ideas of others	○	○	○	○	○	○
I listen to the comments of others	○	○	○	○	○	○
I consider the feedback of others	○	○	○	○	○	○
I reflect upon the comments of others	○	○	○	○	○	○
I observe the strategies of others	○	○	○	○	○	○

	1 - very untrue of me	2	3	4	5	6 - very true of me
I observe how others are doing	○	○	○	○	○	○
I look for confirmation of my understanding from others	○	○	○	○	○	○
I request information from others	○	○	○	○	○	○
I respond to the contributions that others make	○	○	○	○	○	○
I challenge the strategies of others	○	○	○	○	○	○

	1 - very untrue of me	2	3	4	5	6 - very true of me
I challenge the perspectives of others	○	○	○	○	○	○
I help the learning of others	○	○	○	○	○	○
I monitor the learning of others	○	○	○	○	○	○

I feel a sense of belonging in this class.

Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree

Please describe your answer:

I feel supported by my (subject i.e. computer science) peers when I face challenges.
Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree

Please describe your answer to "I feel supported by my (subject i.e. computer science) peers when I face challenges."

Please describe the effects of the weekly reflection groups and reflection questions on yourself:

Please describe the effects of the weekly reflection groups and reflection questions on others in your reflection group:

Please share potential improvements for the weekly reflection groups:

Please share potential improvements for the weekly reflection questions:

Please share potential improvements for the course in general:

What career goals do you have, if any?

Are you currently searching for jobs (this includes searching for a new job if currently employed)?

Please answer the following questions for your most recent job search (for example, if you are currently searching, answer for your current search. If you are currently employed and not also searching for a new position, answer for your prior job search.)

What kind of job or position are/were you looking for?

About how many times for that job search have you:
submitted a job application
been invited to a screening/shorter phone, video, or other interview
been invited to a full job interview
done a job interview
received a job offer
accepted a job offer

Appendix L: Longitudinal follow-up surveys
Please help us learn how to improve the <Course number e.g. COS100> course you took in <Semester Year e.g. Fall 2024>.

Your answers are confidential. There are no right or wrong answers. You may skip any questions you are uncomfortable answering.

Your Email:

Majors, Career

How confident do you feel in your programming abilities? (Scale: 1-7, where 1 is "Not confident at all" and 7 is "Very confident")

How comfortable are you reading, writing, speaking, and listening in English? (Scale: 1-7, where 1 is "Not comfortable at all" and 7 is "Extremely comfortable")

How many hours do you work for pay or at a job, each week, on average?

What jobs and/or careers do you currently have, if any? (Open-ended question)

Part A: If you are still in university or other education/training programs (if not, skip to Part B below):
What are some future careers you are considering after college? (Open-ended question)

What is your intended major(s) and/or minor(s)? (Open-ended question)

What year of university are you in? (Select one: Freshman (First Year), Sophomore (Second Year), Junior (Third Year), Senior (Fourth Year), Graduate Student, Other - please specify)

Part B: If you are no longer taking courses, please answer the following (if not, skip and click the Next button below):
What are some future jobs or careers you are considering after or in addition to your current ones? (Open-ended question)

What major(s) and/or minor(s) did you graduate with and when? (Open-ended question)

What career goals do you have, if any?

Are you currently searching for jobs (this includes searching for a new job if currently employed)?

Please answer the following questions for your most recent job search (for example, if you are currently searching, answer for your current search. If you are currently employed and not also searching for a new position, answer for your prior job search.)

What kind of job or position are/were you looking for?

About how many times for that job search have you:
submitted a job application
been invited to a screening/shorter phone, video, or other interview
been invited to a full job interview
done a job interview
received a job offer

accepted a job offer

Self-efficacy (Likert Scale)

I will be able to achieve most of the goals that I set for myself.

When facing difficult tasks, I am certain that I will accomplish them.

In general, I think that I can obtain outcomes that are important to me.

I believe I can succeed at most any endeavor to which I set my mind.

I will be able to successfully overcome many challenges.

I am confident that I can perform effectively on many different tasks.

Compared to other people, I can do most tasks very well.

Even when things are tough, I can perform quite well.

I generally manage to solve difficult academic problems if I try hard enough.

I know I can stick to my aims and accomplish my goals in my field of study.

I will remain calm in my exam because I know I will have the knowledge to solve the problems.

I know I can pass the exam if I put in enough work during the semester.

The motto ‘If other people can, I can too’ applies to me when it comes to my field of study.

Self-reflection and Insight Scale (Likert)

I don't often think about my thoughts

I rarely spend time in self-reflection

I frequently examine my feelings

I don't really think about why I behave in the way that I do

I frequently take time to reflect on my thoughts

I often think about the way I feel about things

I am not really interested in analyzing my behavior

It is important for me to evaluate the things that I do

I am very interested in examining what I think about

It is important to me to try to understand what my feelings mean

I have a definite need to understand the way that my mind works

It is important to me to be able to understand how my thoughts arise

I am usually aware of my thoughts

I'm often confused about the way that I really feel about things

I usually have a very clear idea about why I've behaved in a certain way

I'm often aware that I'm having a feeling, but I often don't quite know what it is

My behavior often puzzles me

Thinking about my thoughts makes me more confused

Often I find it difficult to make sense of the way I feel about things

I usually know why I feel the way I do

Lifelong learning skills scale (5-point Likert-type scale (from 1 = never to 5 = always)

I am aware of the areas in which I need to learn.
I am aware of the topics that I am more focused on when learning.
I am aware of my strengths in terms of abilities.
I am aware of what I need to learn.
I am aware of the abilities that I need to improve.

I am able to set my own learning goals.
I am capable of planning my own learning strategies.
I am capable of prioritizing my learning needs.
I am capable of setting up my own learning plan.
I am capable of anticipating my learning outcomes.

I create outlines to assist with my understanding of the material.
I take notes on what I am learning.
During lectures, I ask questions to aid in my learning.
I search for learning resources on my own.

While learning, I record questions that arise.

I monitor my learning progress.
I check if I have achieved my learning goals.
I reflect on my learning process.
I take note of the feedback that my friends give me on my learning.
I make revisions based on the feedback I receive from my friends.

I participate in learning groups that involve interpersonal interactions.
I use communication software to communicate with learning partners.
I express my ideas to learning partners verbally.
I take the initiative to discuss with learning partners.
I maintain a good relationship with learning partners.

Self-efficacy

Self-efficacy is a measurement about your confidence in your ability to perform a task. In the following sections, we ask you to evaluate your self-efficacy with regards to programming.

This survey was designed to be broadly applicable. Nevertheless, it uses a number of technical terms, which may or may not be familiar to you depending on the languages or paradigms you might have experienced. In this sense, interpret the phrase “function or method” as “function” or “method” depending on your background. If you do not recognize a term, please select “no answer” for that item.

		strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
1.	I can trace the execution of a program.	☐	☐	☐	☐	☐	☐	☐	☐
2.	I can trace the execution of a program with nested conditionals.	☐	☐	☐	☐	☐	☐	☐	☐
3.	I can trace the execution of a program that uses a list or array.	☐	☐	☐	☐	☐	☐	☐	☐
4.	I can trace the execution of a program that iterates over nested structures such as lists of lists or arrays of arrays.	☐	☐	☐	☐	☐	☐	☐	☐
5.	I can trace the execution of a program that contains function or method calls with arguments and return values.	☐	☐	☐	☐	☐	☐	☐	☐
6.	I can trace the execution of a program that uses the same variable name in different locations.	☐	☐	☐	☐	☐	☐	☐	☐
7.	Given a program, I can explain its purpose in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
8.	I can write a program to solve a problem given the solution to a similar problem.	☐	☐	☐	☐	☐	☐	☐	☐
9.	I can write a program that defines and calls functions or methods.	☐	☐	☐	☐	☐	☐	☐	☐
10.	I can explain the difference between built-in and user-defined types.	☐	☐	☐	☐	☐	☐	☐	☐
11.	I can evaluate boolean expressions I encounter in a program.	☐	☐	☐	☐	☐	☐	☐	☐
12.	I can construct a boolean expression to represent a condition described in plain English.	☐	☐	☐	☐	☐	☐	☐	☐
13.	I can explain the difference between a class and an instance of a class.	☐	☐	☐	☐	☐	☐	☐	☐

	strongly disagree 1	2	3	50/50 4	5	6	strongly agree 7	no answer
14. I can design a set of classes that leverages inheritance where appropriate.	☐	☐	☐	☐	☐	☐	☐	☐
15. Given a class that has two different functions or methods with the same name, I can identify which one was called.	☐	☐	☐	☐	☐	☐	☐	☐
16. I can trace the execution of a program that uses recursion.	☐	☐	☐	☐	☐	☐	☐	☐
17. I can identify recursive patterns in a problem and write a program that exploits this structure.	☐	☐	☐	☐	☐	☐	☐	☐
18. I can write a function that produces a new list/array by processing data from another list or array.	☐	☐	☐	☐	☐	☐	☐	☐
19. Given a program that calls functions or methods, I can describe the state of the system at a particular point in execution.	☐	☐	☐	☐	☐	☐	☐	☐
20. I can decompose a problem into a set of simpler subproblems.	☐	☐	☐	☐	☐	☐	☐	☐
21. I can use a process to analyze a problem, design a solution, and then implement it in code.	☐	☐	☐	☐	☐	☐	☐	☐
22. Given the design of a solution and an incorrect program, I can identify the source of the error.	☐	☐	☐	☐	☐	☐	☐	☐
23. I can design a set of tests that can be used to assess the correctness of a program.	☐	☐	☐	☐	☐	☐	☐	☐
24. I can correctly trace the execution of a program even if my prior attempts to trace similar programs have not succeeded.	☐	☐	☐	☐	☐	☐	☐	☐
25. I can write a correctly functioning program even if I have already spent hours fixing errors and have just found a new error.	☐	☐	☐	☐	☐	☐	☐	☐

If you have and comments about terms used in this survey, please write them here:

(free response textbox for the question above)

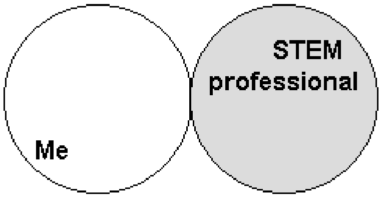
Professional and STEM identity

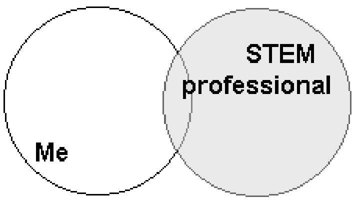
(Likert scale)
When I work with technology, I consider whether there are any related ethical issues that I or my team should address.
I feel equipped to reason about ethical issues arising from technology.
It is important for me to be able to effectively communicate about ethical issues arising from technology.
I would like to gain more experience with recognizing and reflecting upon ethical issues arising from technology.
All computer science students should learn about ethical issues arising from technology as part of their computer science program.
When I work with technology in the future, I expect I will more frequently consider whether there are any related ethical issues that I or my team should address.
I routinely think about ethical issues arising from technology.
I routinely discuss ethical issues raised by technology with my peers, colleagues, or supervisors.
I am interested in learning more about ethical issues arising from technology.

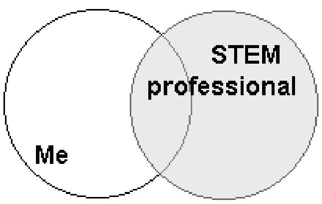
It is important for computer scientists to consider ethical issues arising from technology in their work.
I feel equipped to identify ethical issues arising from technology.
In the future, I expect I will more frequently reflect upon ethical issues arising from technology.
I have discussed or communicated my position on ethical issues arising technology on many occasions.
I routinely use my ability to recognize, reason about, and discuss ethical issues arising from technology.
I am aware of cases where individuals used their technical expertise to help address the ethical issues arising from technology.
Technology is having an impact on society that raises ethical issues.
I have reflected upon ethical issues arising from technology on many occasions.
I have the skills to carry on a meaningful conversation about ethical issues arising from technology.
I want to improve my ability to recognize, reason about, and discuss the ethical issues arising from technology.
If I am faced with ethical issues arising from technology, I will take steps to address them.
I have noticed ethical issues arising from technology.
When reasoning about ethical issues arising from technology, it is important to consider a wide range of perspectives.
I feel empowered to raise ethical issues arising from technology that may arise in my career.
I expect that, in the future, I will pursue opportunities to improve my ability to recognize, reason about, and discuss ethical issues arising from technology.

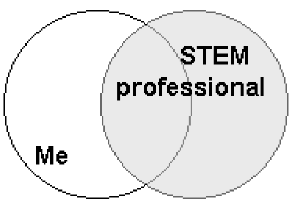
I have the capability to take action in response to ethical issues arising from technology, such as suggesting a design modification or requesting that the issue be brought forward for review.

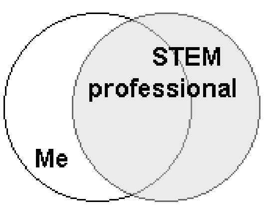
STEM professionals are individuals whose professional activities relate to the STEM fields (Science, Technology, Engineering, or Mathematics). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a STEM professional is.

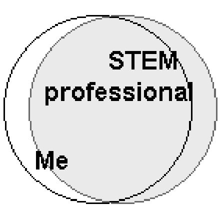












(another copy of the question above but for “Creative professionals are individuals whose professional activities relate to the creative fields (Art, Design, Creative Writing, Music, Film, Photography, and Games). Select the picture that best describes the current overlap of the image you have of yourself and your image of what a Creative professional is.”)

Sense of belonging

1. I feel like I “belong” in computer science.

2. I consider myself as a scientist, technologist, engineer, or mathematician.

3. I take pride in my computing abilities.

on a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree, with a 3 = neutral.

Professional and reflection skills

(Likert scale, 7-point scale from "very inaccurate" to "very accurate.”)
For the teams in this class:

If you make a mistake on this team, it is often held against you.
Members of this team are able to bring up problems and tough issues.
People on this team sometimes reject others for being different.
It is safe to take a risk on this team.
It is difficult to ask other members of this team for help.
No one on this team would deliberately act in a way that undermines my efforts.
Working with members of this team, my unique skills and talents are valued and utilized.

When I am engaged in the learning process as an INDIVIDUAL:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my effort						
I am aware of my thinking						
I know my level of motivation						
I question my thoughts						
I make judgments about the difficulty of a problem						
	1 - very untrue of me	2	3	4	5	6 - very true of me
I am aware of my existing knowledge						
I assess my understanding						
I change my strategy when I need to						
I am aware of my level of learning						
I search for new strategies when needed						
	1 - very untrue of me	2	3	4	5	6 - very true of me
I apply strategies						
I assess how I approach the problem						
I assess my strategies						

When I am engaged in the learning process as a member of a GROUP:

	1 - very untrue of me	2	3	4	5	6 - very true of me
I pay attention to the ideas of others	○	○	○	○	○	○
I listen to the comments of others	○	○	○	○	○	○
I consider the feedback of others	○	○	○	○	○	○
I reflect upon the comments of others	○	○	○	○	○	○
I observe the strategies of others	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I observe how others are doing	○	○	○	○	○	○
I look for confirmation of my understanding from others	○	○	○	○	○	○
I request information from others	○	○	○	○	○	○
I respond to the contributions that others make	○	○	○	○	○	○
I challenge the strategies of others	○	○	○	○	○	○
	1 - very untrue of me	2	3	4	5	6 - very true of me
I challenge the perspectives of others	○	○	○	○	○	○
I help the learning of others	○	○	○	○	○	○
I monitor the learning of others	○	○	○	○	○	○

I feel supported by my (subject i.e. computer science) peers when I face challenges.
Strongly disagree, Somewhat disagree, Neither agree nor disagree, Somewhat agree, Strongly agree

Please describe your answer to "I feel supported by my (subject i.e. computer science) peers when I face challenges."

Please describe the long-term effects on yourself, if any, from the reflection groups and/or weekly reflections from the course?

What practices, if any, have you continued or partly continued from the reflection groups and/or weekly reflections?

If so, around how often do you do each of those practices?

Please share potential improvements for the weekly reflection groups or the course in general:

If there is anything else you would like to add that might be relevant, please do so here.

We value and appreciate your sharing and responses to these questions. Thank you for helping us continue to improve this class, which also helps future students.

Appendix M: Student recruitment script for courses without reflection groups

As a part of this class, you have the opportunity to participate in research on how different methods of teaching and assignments affect learning in a course like this one. Participation in the research is voluntary and will not affect your assignments, class activities, or your grade in any way. I will not know who participated in the study until after grades are submitted.

You can participate by allowing the researchers to analyze your work in the class. You may also choose to participate in additional confidential follow-up surveys after the course, which will take about 10-15 minutes each, roughly every six months, for four years (8 surveys total). For each longitudinal survey you submit, you will receive a $10 VISA gift card.

More details are in the consent form. We'll take some time now in class to read the consent form and you may follow the directions there to opt-out if you wish, by emailing one of the researchers who is not the instructor of this course, such as <other person on the research team who is not the instructor>. You may decide now or wait until the end of the week. We encourage you to participate as it will help us improve the course, but it is completely optional and will have no effect on your grade. You must be at least 18 years old to participate. If you have any questions, contact <other person on the research team who is not the instructor>, another faculty member involved in the research, whose email is on the consent form and is also here to answer any questions you have.